

KZ-Based Spectral Modeling of Nonlinear Interference with Data-Driven PSD Estimation

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Abstract Accurate modeling of nonlinear interference in optical fiber systems is investigated using Kolmogorov–Zakharov modeling and learning-based PSD estimation. KZ outperforms GN in accuracy and effective complexity. A linear regression model maps spectra to NLI PSD, enabling improved performance prediction in practical systems. ©2026 The Author(s)

Introduction

Accurate modeling of nonlinear interference (NLI) is critical for predicting and optimizing coherent wavelength-division multiplexed (WDM) optical systems, where Kerr-induced distortions limit performance. While the split-step Fourier (SSF) method provides accurate solutions, its high computational cost motivates analytical models such as Gaussian noise (GN) [1, 2] and enhanced-GN (EGN) [3–5]. The Kolmogorov–Zakharov (KZ) model has recently emerged as an alternative spectral framework with promising accuracy [6–8]. However, its performance has not been systematically evaluated across practical scenarios, and existing studies focus on received power spectral density (PSD) rather than the nonlinear-distortion PSD governing system performance. In this paper, we show that KZ can outperform GN in PSD prediction accuracy while maintaining favorable complexity. Furthermore, we introduce a learning-assisted estimator that maps input and output PSDs to NLI PSD, enabling unified and system-level comparison of analytical models against SSF simulations.

Stochastic Models Comparison: KZ vs GN

The EGN model is widely used for predicting NLI in optical fiber systems; however, it relies on the assumption that NLI is additive and statistically independent of the signal. This leads to a fundamental inconsistency: in an ideal lossless and noise-free system, total signal energy must be conserved, yet the EGN formulation predicts strictly additive NLI power across all frequency components, implying an artificial increase in total energy. In contrast, the KZ model, derived from a wave-turbulence framework, inherently preserves energy by modeling nonlinear propagation as a redistribution of spectral power across frequencies, rather than as an additive noise term.

This distinction has important implications for PSD prediction. Fig. 1 compares the spectral evolution predicted by the KZ [8, Eq. 45] and EGN [4, Eq. 5] models against SSF simulations for a five-channel WDM system with 64-QAM modulation at 40 GBaud per channel, 50 GHz channel

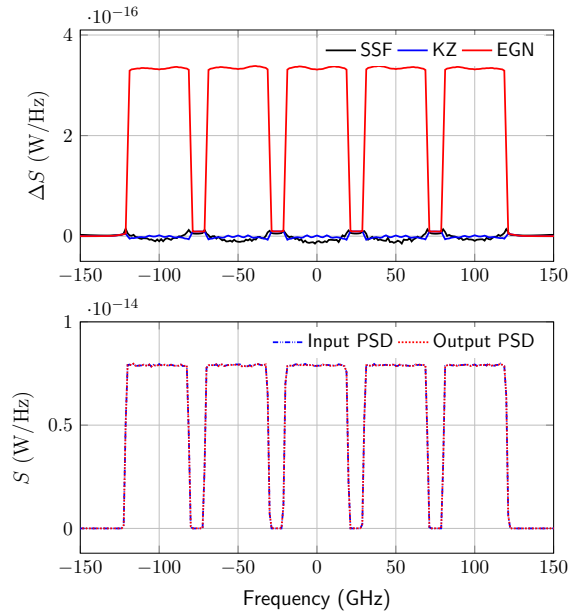


Fig. 1: Top: $\Delta S = S_{out} - S_{in}$ for the KZ and EGN models compared to the SSF method. Bottom: corresponding input and output PSDs. The launch power is -5 dBm per channel and the transmission distance is 2400 km.

spacing, and root-raised-cosine pulses with roll-off 0.1. Transmission is performed over 30 spans of 80 km standard single-mode fiber (2400 km total). SSF-based Monte Carlo simulations are carried out over 10,000 realizations, and output PSDs are obtained via ensemble averaging to serve as the reference. The results show that while the EGN model overestimates the spectral deviation, the KZ model closely matches the SSF reference around the optimal launch power. Let S_{in} and S_{out} denote the input and output PSDs, and define $\Delta S \triangleq S_{out} - S_{in}$. Quantitatively, the KZ prediction satisfies

$$|\Delta S_{SSF} - \Delta S_{KZ}| < |\Delta S_{SSF} - \Delta S_{EGN}|,$$

demonstrating improved accuracy in capturing nonlinear spectral distortions. The top plot shows ΔS , while the bottom plot shows the corresponding input and output PSDs, allowing visual assessment of the relative magnitude of the nonlinear distortions.

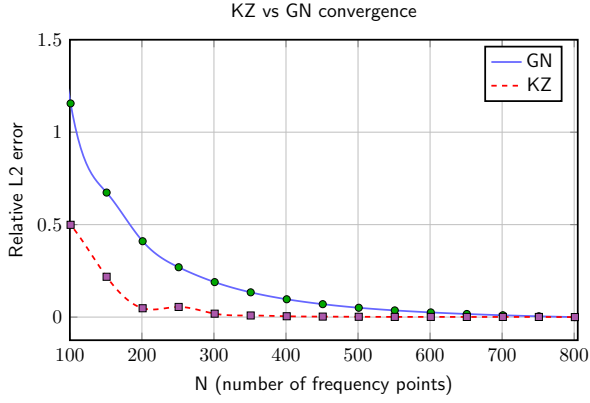


Fig. 2: Relative approximation error of GN and KZ spectra vs. high-resolution reference (40 GHz, 0 dBm, 2400 km).

It is important to note that, unlike EGN, the KZ model does not interpret ΔS as an independent NLI PSD, since signal–nonlinearity correlations are preserved. Instead, it reflects the underlying spectral energy transfer induced by nonlinear interactions, leading to a more physically consistent description of NLI where energy is redistributed across frequencies rather than simply added, explaining the improved agreement with SSF simulations. The KZ model tends to exhibit greater accuracy than EGN in scenarios where nonlinear effects dominate.

Computational Complexity

From the formulation of the KZ model in [8, Eq. 45], it can be observed that the evaluation at each frequency component involves a double summation over frequency indices, resulting in a computational complexity of $\mathcal{O}(N^2)$ per frequency point. Similarly, the GN and EGN models have also been shown to scale quadratically with the number of frequency components [4].

While all models share the same asymptotic $\mathcal{O}(N^2)$ complexity, a structural difference arises in the number of contributing terms. In particular, the KZ model consists of multiple quadratic terms, whereas the GN/EGN formulation involves a single such term, leading to a somewhat higher per-evaluation constant for KZ. However, since the asymptotic scaling is identical, this difference alone does not fully determine practical computational efficiency.

To assess practical complexity, we perform a convergence study in which each model is evaluated over increasingly refined frequency grids. For each grid size N , the predicted PSD $\hat{\mathbf{x}}_N$ is compared against a high-resolution reference $\hat{\mathbf{x}}_{N_{\max}}$ with $N_{\max} = 801$, using the relative ℓ_2 error

$$\varepsilon_2(N) = \frac{\|\hat{\mathbf{x}}_N - \hat{\mathbf{x}}_{N_{\max}}\|_2}{\|\hat{\mathbf{x}}_{N_{\max}}\|_2}.$$

Fig. 2 shows that the KZ model converges at $N \approx 300$, whereas the GN model requires $N \approx 700$ samples to reach a similar error level.

To quantify the total computational effort, let $W(N)$ denote the number of real-valued operations required for a grid of size N . Since both models scale quadratically, we write $W_{\text{GN}}(N) \approx C_{\text{GN}}N^2$ and $W_{\text{KZ}}(N) \approx C_{\text{KZ}}N^2$, with constants $C_{\text{GN}} \approx 4.64$ and $C_{\text{KZ}} \approx 11.03$, obtained via operation-count analysis. Evaluating at the respective convergence points yields $W_{\text{KZ}}(300) \approx 9.93 \times 10^5$ and $W_{\text{GN}}(700) \approx 2.27 \times 10^6$. Thus, despite a larger per-evaluation constant, the KZ model achieves its asymptotic accuracy at a smaller grid size, resulting in an overall computational cost that is approximately $2.3\times$ lower than that of the GN model for comparable accuracy.

Learning-Based Estimation

In this section, we use data-driven methods to estimate the NLI error PSD at the output of the fiber. The goal is to construct a fast surrogate that maps spectral observations to nonlinear distortion.

We consider the problem of estimating the NLI PSD from spectral observations of an optical link. Specifically, given the input PSD $\mathbf{x}$ and the output PSD $\mathbf{y}$, both sampled on a common frequency grid, we aim to predict the corresponding NLI PSD $\mathbf{z}$. This defines a supervised regression task of the form $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{z}$, where all quantities are real-valued vectors of length N . In addition to spectral inputs, the estimator may optionally incorporate a small set of scalar system parameters μ —such as roll-off factor, symbol rate, launch power, and propagation distance—leading to the extended mapping $(\mathbf{x}, \mathbf{y}, \mu) \mapsto \mathbf{z}$. In this work, we consider both formulations. The motivation for this approach is that $\mathbf{z}$ directly determines system performance metrics such as SNR, while $\mathbf{x}$ and $\mathbf{y}$ are readily available from either simulations or analytical models. A reliable estimator of these mappings therefore provides a fast surrogate for NLI prediction, enabling efficient evaluation across a wide range of system configurations.

The training data consists of approximately 231,000 samples generated via SSF simulations over a wide range of transmission parameters. Each sample contains the triplet $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in \mathbb{R}^{128}$, representing the input PSD, output PSD, and corresponding NLI PSD, all defined on a common frequency grid. The model input is constructed as the concatenation $\mathbf{u} = [\mathbf{x}; \mathbf{y}] \in \mathbb{R}^{256}$, with optional inclusion of scalar parameters, i.e., μ . The dataset is randomly split into training, validation, and test sets (70%/20%/10%), and the same split is used for all models.

To address the large dynamic range of PSD values, log-domain transformations are applied elementwise as $\log(\cdot + \varepsilon)$ with $\varepsilon = 10^{-40}$ to $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$, with predictions mapped back via the exponential function. This improves numerical conditioning and approximately linearizes multiplicative scaling

Tab. 1: Validation performance of linear and structured linear models (mean relative error in %).

Mean Relative Error (%)		
Model	no meta	meta
Linear (raw)	4.3×10^5	7.6×10^5
Linear (log)	23	18
PCA(10)+Poly(4)+Ridge	11.5	6
PCA(80)+Poly(2)+Ridge	6.2	4.3
PCA(35)+Poly(3)+Ridge	3.6	2.7
PCA(20)+Poly(4)+Ridge	3.2	2.4

effects. In addition, ridge regularization [9, 10] is employed in the linear models to improve stability and mitigate overfitting.

To exploit the strong correlation across frequency bins, principal component analysis (PCA) is optionally applied after standardization, reducing the input dimension while preserving dominant spectral structure. In addition, polynomial feature expansion is used in conjunction with ridge regression to capture nonlinear dependencies while maintaining a linear model structure.

The models are evaluated using the mean relative ℓ_2 error. Let $\mathbf{z}_n$ and $\hat{\mathbf{z}}_n$ denote the true and predicted NLI PSDs for test sample n , and let N_{test} be the number of test samples. The evaluation metric is defined as

$$\epsilon_{\text{mean}} = \frac{1}{N_{\text{test}}} \sum_{n=1}^{N_{\text{test}}} \frac{\|\hat{\mathbf{z}}_n - \mathbf{z}_n\|_2}{\|\mathbf{z}_n\|_2 + \epsilon_{\text{rel}}}, \quad \epsilon_{\text{rel}} = 10^{-40}.$$

A broad range of linear models was trained to estimate the NLI PSD, including raw-domain linear regression, log-domain linear and ridge regression, and PCA-reduced polynomial ridge models. In all models except the raw baseline, both the input feature vector $\mathbf{X} = [\mathbf{x}, \mathbf{y}]$ and the target $\mathbf{z}$ were represented in the log domain, and the predicted $\hat{\mathbf{z}}$ is mapped back to the original domain via the exponential function. Table 1 reports a representative subset of these models and their relative errors. Although the PCA(20)+Poly(4)+Ridge model achieves the lowest mean relative error, additional sensitivity analysis indicates that it is more prone to overfitting. This is consistent with the low-rank structure of the underlying PSD data, where most of the useful variance is concentrated in roughly the first ~ 10 principal components, while higher components primarily capture weak or noisy directions. For this reason, the PCA(10)+Poly(4)+Ridge model is selected as the best linear model overall, since it provides slightly higher mean error but more stable and physically meaningful estimation.

Fig. 3 illustrates the NLI-PSD estimation obtained using the selected linear model, PCA(10)+Poly(4)+Ridge in the log domain, for both GN- and KZ-based spectral inputs, with SSF serving as the reference. The results show that

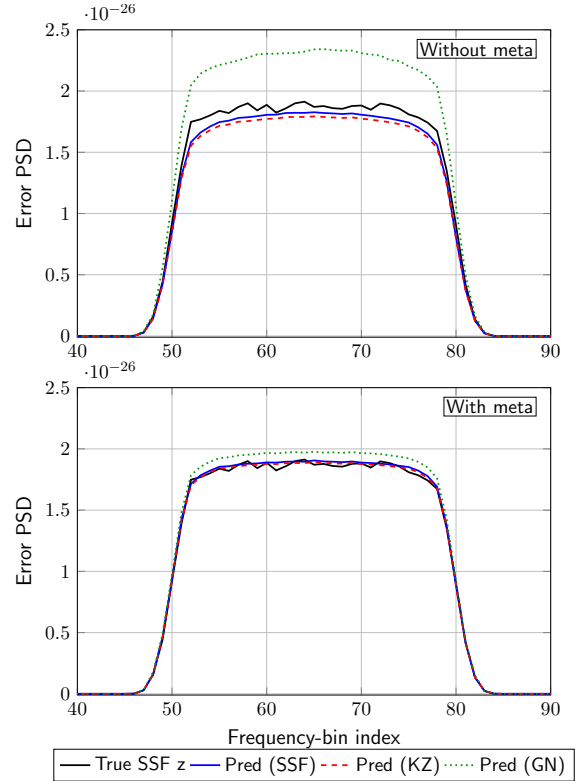


Fig. 3: Estimated NLI PSD using PCA(10)+Poly(4)+Ridge (log domain) for GN and KZ inputs, compared to SSF. Top: without metadata. Bottom: with metadata. Shown over the active spectral region.

the KZ-based predictions closely follow the SSF curve, while the GN-based predictions exhibit noticeable deviation. Quantitatively, the relative error is lower for KZ (6.35%) compared to GN (22.37%), and further improves when metadata is included, reaching 1.34% for KZ and 5.15% for GN. This behavior is consistent across simulations, and the figure shows a representative example. For clarity, the figure is zoomed to the active spectral region. These results demonstrate that the proposed learning-based framework provides an accurate NLI PSD estimate, enabling improved prediction of system-level metrics such as SNR and overall performance. The consistent advantage of KZ-based inputs further highlights their superior fidelity in capturing nonlinear spectral behavior.

Conclusions

This paper presented a unified framework for evaluating NLI in optical fiber systems by combining KZ-based modeling with learning-assisted PSD estimation. The results show that KZ provides more accurate spectral predictions than GN while achieving lower effective computational cost. The proposed learning-based estimator enables direct NLI PSD evaluation from spectral inputs, linking analytical models to system-level performance. Together, these results demonstrate an efficient and physically consistent approach for improved NLI modeling and performance prediction.

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